

5.1 特征值与特征向量

A 为 n 阶方阵. 对于数 λ . 若存在非 0 列向量 α . 使 $A\alpha = \lambda\alpha$. λ 称为一个特征值. α 为对应 λ 的特征向量

注:

λ 可以为 0. 特征向量不能为 0

$$A\alpha = \lambda\alpha \quad \therefore \lambda\alpha - A\alpha = 0$$

$$(\lambda E - A)\alpha = 0 \quad \because \alpha \text{ 为非零列向量}$$

即求 $(\lambda E - A)X = 0$ 的非零解.

$$\Leftrightarrow \text{系数行列式为 } 0 \quad |\lambda E - A| = 0$$

定义: $\lambda E - A$ 为特征矩阵.

$|\lambda E - A|$ 为特征多项式.

$$|\lambda E - A| = 0 \quad \text{特征方程.}$$

解出特征方程 λ 的根即为所求特征值(根)

1) λ 是 A 的特征值, α 是 λ 对应的特征向量, $c\alpha$ 也是 λ 对应的特征向量 ($c \neq 0$)

$$A\alpha = \lambda\alpha \quad cA\alpha = c\lambda\alpha$$

$$A(c\alpha) = \lambda(c\alpha)$$

对应 λ 的特征向量不是唯一的
但一个特征向量 α 只能对应一个特征值

若 α ($\alpha \neq 0$) 是 λ_1, λ_2 ($\lambda_1 \neq \lambda_2$) 的特征向量

$$A\alpha = \lambda_1\alpha \quad A\alpha = \lambda_2\alpha$$

$$\lambda_1\alpha = \lambda_2\alpha \quad \cancel{\lambda_1\alpha}$$

$$(\lambda_1 - \lambda_2)\alpha = 0 \quad \alpha = 0 \text{ 矛盾}$$

2) 若 α_1, α_2 是 λ 的特征向量.

$c_1\alpha_1 + c_2\alpha_2$ 是 λ 的特征向量.

$$A(c_1\alpha_1 + c_2\alpha_2) = c_1A\alpha_1 + c_2A\alpha_2$$

$$A\alpha_1 = \lambda\alpha_1 \quad A\alpha_2 = \lambda\alpha_2$$

$$\begin{aligned} c_1A\alpha_1 + c_2A\alpha_2 &= c_1\lambda\alpha_1 + c_2\lambda\alpha_2 \\ &= \lambda(c_1\alpha_1 + c_2\alpha_2) \end{aligned}$$

$$\text{即 } A(c_1\alpha_1 + c_2\alpha_2) = \lambda(c_1\alpha_1 + c_2\alpha_2)$$

$$\text{例: } A = \begin{bmatrix} -1 & 1 & 0 \\ -4 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix} \text{ 的特征值.}$$

$$\begin{aligned} \text{解: } \lambda E - A &= \begin{bmatrix} \lambda & & \\ & \lambda & \\ & & \lambda \end{bmatrix} - \begin{bmatrix} -1 & 1 & 0 \\ -4 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} \lambda+1 & -1 & 0 \\ 4 & \lambda-3 & 0 \\ -1 & 0 & \lambda-2 \end{bmatrix} \end{aligned}$$

$$|\lambda E - A| = \begin{vmatrix} \lambda+1 & -1 & 0 \\ 4 & \lambda-3 & 0 \\ -1 & 0 & \lambda-2 \end{vmatrix}$$



如何求解?

1) 完全展开, 最好不要, 将得到一个三次的方程.

2) 把某行尽可能转化为0, 按行展开

$$\text{如 } (\lambda-2)(-1)^{3+3} \begin{vmatrix} \lambda+1 & -1 \\ 4 & \lambda-3 \end{vmatrix}$$

3) 提公因子 (含有 λ)

4) 相反数, 相同数, 行和/列和
继续求解该题

$$|\lambda E - A| = (\lambda-2)(\lambda-1)(\lambda-1)$$

$$\lambda_1 = \lambda_2 = 1 \quad \lambda_3 = 2$$

(必有实根三个根, 重复的需写成重根)

$$(\lambda E - A)x = 0 \xrightarrow{x \neq 0} (\lambda E - A)x = 0 \text{ 非零解}$$

1) $\lambda_1 = \lambda_2 = 1$ 时.

$$\lambda E - A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & -2 & 0 \\ -1 & 0 & -1 \end{bmatrix} \text{ 基础解系为 } \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$\Rightarrow \text{全部特征向量 } c_1 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \quad c_1 \neq 0$$

$$2) \lambda_3 = 2 \quad \lambda E - A = \begin{bmatrix} 3 & -1 & 0 \\ 4 & -1 & 0 \\ -1 & 0 & 2 \end{bmatrix}$$

$$\text{基础解系 } \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{特征向量 } c_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad c_2 \neq 0$$

$$\text{例 2. } A = \begin{bmatrix} 1 & -2 & 2 \\ -2 & -2 & 4 \\ 2 & 4 & -2 \end{bmatrix}$$

注: $|\lambda E - A| = 0$ λ 只在主对角线存在

② A 所有元素都取相反数

$$|\lambda E - A| = \begin{vmatrix} \lambda+1 & 2 & -2 \\ 2 & \lambda+2 & -4 \\ -2 & -4 & \lambda+2 \end{vmatrix} \quad \text{验证!}$$

$$= \begin{vmatrix} \lambda+1 & 2 & -2 \\ 2 & \lambda+2 & -4 \\ 0 & \lambda-2 & \lambda-2 \end{vmatrix}$$

$$= (\lambda-2)(\lambda-2)(\lambda+7)$$

$$\lambda_1 = -7 \quad \lambda_2 = \lambda_3 = 2$$

1) $\lambda_1 = -7$

$$\lambda E - A = \begin{bmatrix} -8 & 2 & -2 \\ 2 & -5 & -4 \\ -2 & -4 & -5 \end{bmatrix} \rightarrow \text{行简化}$$

写出同解方程组, 自由未知量, 依次赋值.

$$c_1 \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} \quad c_1 \neq 0$$

同理, $\lambda_2 = \lambda_3 = 2$

$$c_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \quad (c_2, c_3 \text{ 不同时为 } 0)$$



如何解?

1) 完全展开, 最好不要, 将得到一个三次的方程.

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$$\text{如 } (\lambda-2)(-1)^{3+3} \begin{vmatrix} \lambda+1 & -1 \\ 4 & \lambda-3 \end{vmatrix}$$

3) 提公因子 (含有 λ)

4) 相反数, 相同数, 行和/列和

继续求解该题

$$|\lambda E - A| = (\lambda-2)(\lambda-1)(\lambda-1) \\ (\lambda-2)(\lambda-1)(\lambda-1)$$

$$\lambda_1 = \lambda_2 = 1 \quad \lambda_3 = 2$$

(必有实根三个根, 重复的需写成重根)

$$(\lambda E - A)x = 0 \xrightarrow{x \neq 0} (\lambda E - A)x = 0 \text{ 非零解}$$

1) $\lambda_1 = \lambda_2 = 1$ 时.

$$\lambda E - A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & -2 & 0 \\ -1 & 0 & -1 \end{bmatrix} \text{ 基础解系 为 } \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$\Rightarrow \text{全部特征向量 } c_1 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \quad c_1 \neq 0$$

$$2) \lambda_3 = 2 \quad \lambda E - A = \begin{bmatrix} 3 & -1 & 0 \\ 4 & -1 & 0 \\ -1 & 0 & 2 \end{bmatrix}$$

$$\text{基础解系 } \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{特征向量 } c_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad c_2 \neq 0$$

$$\text{例 2. } A = \begin{bmatrix} 1 & -2 & 2 \\ -2 & -2 & 4 \\ 2 & 4 & -2 \end{bmatrix}$$

注: $|\lambda E - A| = 0$ λ 只在主对角线存在

② A 所有元素都取相反数

$$|\lambda E - A| = \begin{vmatrix} \lambda+1 & 2 & -2 \\ 2 & \lambda+2 & -4 \\ -2 & -4 & \lambda+2 \end{vmatrix} \quad \text{检查!}$$

$$= \begin{vmatrix} \lambda+1 & 2 & -2 \\ 2 & \lambda+2 & -4 \\ 0 & \lambda-2 & \lambda-2 \end{vmatrix}$$

$$= (\lambda-2)(\lambda-2)(\lambda+7)$$

$$\lambda_1 = -7 \quad \lambda_2 = \lambda_3 = 2$$

1) $\lambda_1 = -7$

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写出同解方程组, 自由未知量, 依次赋值.

$$c_1 \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} \quad c_1 \neq 0$$

同理, $\lambda_2 = \lambda_3 = 2$

$$c_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \quad (c_2, c_3 \text{ 不同时为 } 0)$$



$$A\alpha = \lambda\alpha$$

$$\lambda\alpha = A\alpha$$

$$A\alpha - \lambda\alpha = 0$$

$$(\lambda E - A)\alpha = 0$$

$$(A - \lambda E)\alpha = 0$$

↓

$$|A - \lambda E| = 0$$

$$|\lambda E - A| = 0$$

$|A - \lambda E| = (-1)^n |\lambda E - A|$ 但对结果无影响. 显然 $|A - \lambda E|$ 更简单.

n 阶三角形矩阵, 特征值是主对角线上的 n 个元素

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ & a_{22} & \cdots & a_{2n} \\ & & \cdots & \\ & & & a_{nn} \end{bmatrix}$$

基础解系 $\rightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$$A\alpha = \lambda\alpha$$

$$0\alpha = 0\alpha \quad \text{对任何非零}\alpha\text{均成立.}$$

$$|\lambda E - A| = \begin{vmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ & \lambda - a_{22} & \cdots & -a_{2n} \\ & & \cdots & \\ & & & \lambda - a_{nn} \end{vmatrix}$$

$$= (\lambda - a_{11})(\lambda - a_{22}) \cdots (\lambda - a_{nn})$$

$$\lambda_1 = a_{11} \quad \lambda_2 = a_{22} \quad \cdots \quad \lambda_n = a_{nn}$$

练习: 求 $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ 的特征值与特征向量.

$$|\lambda E - A| = \begin{vmatrix} \lambda & & \\ & \lambda & \\ & & \lambda \end{vmatrix} = \lambda^3 = 0$$

$$\lambda = 0, (\lambda E - A)x = 0$$

$$\lambda E - A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

同解方程组

$$\begin{cases} 0x_1 + 0x_2 + 0x_3 = 0 \\ \hline \hline \end{cases} \quad \text{全为自由未知量}$$

特征值与特征向量的基本性质

性质 1) A 和 A^T 有相同的特征值

$$\begin{aligned} |\lambda E - A^T| &= |\lambda E^T - A^T| \\ &= |(\lambda E - A)^T| \\ &= |\lambda E - A| \end{aligned}$$

注: 两者特征向量未必相同.

2) 对于一个矩阵 A , 如果

$$\textcircled{1} \sum |a_{ij}| < 1 \quad i = 1, 2, \dots, n$$

$$\textcircled{2} \sum |a_{ij}| < 1 \quad j = 1, \dots, n$$

$$\text{即 } |\lambda_k| < 1$$



即 A 的每行/列元素绝对值之和 < 1
 则特征值的模 < 1

3) 已知矩阵 A 的 n 个特征值.

$$\lambda_1, \lambda_2, \dots, \lambda_n$$

$$\textcircled{1} \text{ 所有 } \sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii}$$

$$\textcircled{2} \lambda_1 \lambda_2 \dots \lambda_n = |A|$$

引例 1) $ax^3 + bx^2 + cx + d$ 的根为
 1, 2, 3.

$$(x-1)(x-2)(x-3) = 0$$

$$\begin{aligned} 2) (x+c_1)(x+c_2) \dots (x+c_n) \\ = x^n + (c_1+c_2+\dots+c_n)x^{n-1} + \dots \\ + c_1 c_2 \dots c_n \end{aligned}$$

$$\text{证: } |\lambda E - A| = \begin{vmatrix} \lambda - a_{11} & -a_{12} & \dots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \dots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & \dots & \dots & \lambda - a_{nn} \end{vmatrix}$$

其中一项为 $(\lambda - a_{11})(\lambda - a_{22}) \dots (\lambda - a_{nn})$
 $= \lambda^n - (a_{11} + \dots + a_{nn})\lambda^{n-1} + \dots + (-1)^n |A|$

$$\lambda = 0 \quad (\lambda E - A) = -A$$

$$\therefore |\lambda E - A| = \lambda^n - (a_{11} + \dots + a_{nn})\lambda^{n-1} + \dots + (-1)^n |A|$$

又特征根为 $\lambda_1, \lambda_2, \dots, \lambda_n$

$$\therefore |\lambda E - A| = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n)$$

$$= (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n)$$

$$= \lambda^n - (\lambda_1 + \lambda_2 + \dots + \lambda_n)\lambda^{n-1}$$

$$+ (-1)^n \lambda_1 \lambda_2 \dots \lambda_n$$

系数对应相等

$$\therefore \lambda_1 + \lambda_2 + \dots + \lambda_n = a_{11} + a_{22} + \dots + a_{nn}$$

$$\text{即 } \sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii}$$

$$(-1)^n \lambda_1 \lambda_2 \dots \lambda_n = (-1)^n |A|$$

$$\text{即 } \lambda_1 \lambda_2 \dots \lambda_n = |A|$$

$\sum_{i=1}^n a_{ii}$ 为矩阵的迹, 记作 $\text{tr}(A)$

若有 $\lambda_i = 0$ 则 $|A| = \lambda_1 \lambda_2 \dots \lambda_n = 0$

则 A 不可逆. 故若要 A 可逆, 需
 所有特征根均不为 0

性质 4) n 阶方阵 A 互不相同的

特征值 $\lambda_1, \lambda_2, \dots, \lambda_m$ 对应的

特征向量 $\alpha_1, \alpha_2, \dots, \alpha_m$ 线性无关.

$m=1$ $\alpha_1 \neq 0$, 线性无关.

$\alpha_1, \dots, \alpha_{s-1}$ 成立, 下证对 s 成立.



扫描全能王 创建

设 $k_1\alpha_1 + k_2\alpha_2 + \dots + k_s\alpha_s = 0$ ① 左乘 A

$$\begin{cases} k_1 A\alpha_1 + k_2 A\alpha_2 + \dots + k_s A\alpha_s = 0 \\ k_1 \lambda_1 \alpha_1 + k_2 \lambda_2 \alpha_2 + \dots + k_s \lambda_s \alpha_s = 0 \\ k_1 \lambda_s \alpha_1 + k_2 \lambda_s \alpha_2 + \dots + k_s \lambda_s \alpha_s = 0 \end{cases} \xrightarrow{\text{相减}}$$

$$k_1(\lambda_1 - \lambda_s)\alpha_1 + k_2(\lambda_2 - \lambda_s)\alpha_2 + \dots + k_{s-1}(\lambda_{s-1} - \lambda_s)\alpha_{s-1} = 0$$

$\therefore$ 对 $s-1$ 成立. 即 $\alpha_1, \alpha_2, \dots, \alpha_{s-1}$ 线性无关

$$\therefore \lambda_1 - \lambda_s = \lambda_2 - \lambda_s = \dots = \lambda_{s-1} - \lambda_s = 0$$

$\therefore$ 特征值互不相同 $\therefore \lambda_i - \lambda_s \neq 0$

$$\therefore k_1 = k_2 = \dots = k_{s-1} = 0$$

$$\text{代入①得 } k_s \alpha_s = 0 \xrightarrow{\alpha_s \neq 0} k_s = 0$$

$\therefore$ 线性无关.

$$\text{例: } A \text{ 的特征值 } \lambda_1=1 \quad \lambda_2=2 \quad \lambda_3=-3$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\alpha_1 \quad \alpha_2 \quad \alpha_3$$

$\alpha_1, \alpha_2, \alpha_3$ 线性无关.

性质5) 设 $\lambda_1, \lambda_2, \dots, \lambda_s$ 是 A 的 s 个互异的特征值. $\xi_1, \xi_2, \dots, \xi_s$ 和 $\eta_1, \eta_2, \dots, \eta_t$

分别是对应于 $\lambda_1, \lambda_2, \dots, \lambda_s$ 的线性无关的特征向量. 其他性质: λ 是 A 的特征值.

1) $k\lambda$ 是 kA 的特征值.

$\eta_1, \eta_2, \dots, \eta_t$ 线性无关.

2) λ^2 是 A^2 的特征值

$$A\alpha = \lambda\alpha \quad (kA)\alpha = (k\lambda)\alpha$$

3) λ^k 是 A^k 的特征值

$$A\alpha = \lambda\alpha \quad A^2\alpha = \lambda A\alpha = \lambda^2\alpha$$

λ^k 是 A^k 的特征值.

$$\text{例: } \lambda_1=1 \quad \lambda_2=2 \quad \lambda_3=-3$$

$$\underbrace{\alpha_1 \alpha_2}_{\text{无关}} \quad \underbrace{\alpha_3}_{\text{无关}} \quad \underbrace{\alpha_4 \alpha_5 \alpha_6}_{\text{无关}}$$

$\Rightarrow \alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6$ 线性无关.

性质6: λ 是 A 的特征多项式的 k 重特征值, A 的对应 λ 的线性无关特征向量最多是 k 个

特别地, 若 λ 是 A 的单根, λ 对应的无关特征向量最多一个.

k 重特征根对应的线性无关的特征向量的个数小于等于 k .

例: A 为 6 阶方阵. $\lambda=1$ (3重)

$\lambda=0$ (2重) $\lambda=5$ (单根)

α_4, α_5

α_1

n 阶矩阵 A 所有线性无关的特征向量的个数最多为 n .



3)

$\varphi(\lambda)$ 是 $\varphi(A)$ 的特征值, 其中

$\varphi(\lambda) = a_0 + a_1\lambda + \dots + a_m\lambda^m$ 是 λ 的多项式 A 的特征根为 -3 .

$\varphi(A) = a_0E + a_1A + \dots + a_mA^m$ 是矩阵

A 的多项式

例: 2 是 A 的特征值

$$\begin{array}{cccc} \text{— 是 } & A^5 & + 6A^2 & + A + 3E \\ & \downarrow & \downarrow & \downarrow \downarrow \\ & 2 & 2 & 2 \quad 1 \end{array}$$

4) λ 是 A 的特征值, $\frac{1}{\lambda}$ 是 A^{-1} 的特征值.

$\frac{1}{\lambda}|A|$ 是 A^* 的特征值.

$$A\alpha = \lambda\alpha \xrightarrow{\text{左乘 } A^{-1}} \alpha = \lambda A^{-1}\alpha$$

$$\therefore A^{-1}\alpha = \frac{1}{\lambda}\alpha$$

$$A^{-1} = \frac{1}{|A|}A^* \therefore \frac{1}{|A|}A^*\alpha = \frac{1}{\lambda}\alpha$$

$$\Rightarrow A^*\alpha = \frac{|A|}{\lambda}\alpha$$

$$\lambda|A|^{n-2}$$

是 $(A^*)^*$ 的特征值

$$\frac{1}{\lambda|A|}|A^*| = \lambda \frac{1}{|A|}|A|^{n-1} = \lambda|A|^{n-2}$$

例: A 是四阶方阵, $|3E+A|=0$

$AA^T=2E$, $|A|<0$, 求 A^* 的一个特

征值

$$|\lambda E - A| = 0$$

$$\text{解: } | -(-3E - A) | = (-1)^4 | -3E - A | = 0$$

$$\left(\begin{array}{l} |E + 2A| = 0 \Rightarrow | -2(-\frac{1}{2}E - A) | = 0 \\ \lambda = -\frac{1}{2} \\ |A| = 0 \quad (-1)^n | 0E - A | = 0 \quad \underline{\lambda = 0} \end{array} \right)$$

$$AA^T = 2E \quad |A||A^T| = |2E| = 2^4$$

$$|A| = |A^T| \therefore |A| = \pm 4$$

$$\because |A| < 0 \therefore |A| = -4$$

$$\frac{1}{\lambda}|A| = -\frac{1}{2} \times (-4) = \frac{4}{2}$$

$$\text{例: } A = \begin{pmatrix} 1 & -3 & 3 \\ 3 & a & 3 \\ 6 & -6 & b \end{pmatrix} \quad \lambda_1 = -2, \lambda_2 = 4, \lambda_3$$

求 a, b, λ 各取何

$$-2 + 4 + \lambda_3 = 1 + a + b$$

$$-2 \times 4 \times \lambda_3 = \begin{vmatrix} 1 & -3 & 3 \\ 3 & a & 3 \\ 6 & -6 & b \end{vmatrix}$$

$$|\lambda_1 E - A| = \begin{vmatrix} -3 & 3 & -3 \\ 3 & -2-a-3 \\ -6 & 6 & -2-b \end{vmatrix}$$

$$= 3(5+a)(4-b) = 0$$



扫描全能王 创建

三阶方阵A的特征值为4, -1, -1

$$|A| = 4 \times (-1) \times (-1) = 4$$

$$\text{tr}(A) = 4 - 1 - 1 = 2$$

例 若 $\lambda = 2$ 为一个特征根^① 求 $|E + 2A|$ 的

特征根^② $|A^T|$

① $E \rightarrow I \quad A \rightarrow \lambda \quad 1+2 \times 2 = 5$

② $|A^T|$ 与 $|A|$ 同, 为 4

③ $A^{-1} \rightarrow \frac{1}{\lambda}$

④ $A^* \rightarrow \frac{1}{\lambda} |A| = \frac{k}{\lambda}$

5.2 相似矩阵. 可对角化条件.

相似: A, B 为 2 个 n 阶方阵.

若存在 n 阶可逆 P , 使 $P^{-1}AP = B$

则记作 $A \sim B$

性质

1) 反身性 $A \sim A \quad E^{-1}AE = A$

2) 对称性 $A \sim B$, 则 $B \sim A$

3) 传递性 $A \sim B, B \sim C \Rightarrow A \sim C$

$$\because A \sim B \quad \therefore P^{-1}AP = B$$

$$\because B \sim C \quad \therefore Q^{-1}BQ = C$$

$$\therefore Q^{-1}P^{-1}APQ = C$$

$$(PQ)^{-1}APQ = C$$

性质

1) 若 $A \sim B$, A, B 有不相同的

特征值, $|A| = |B|, \text{tr}(A) = \text{tr}(B)$

$$|\lambda E - A| \quad P^{-1}AP = B$$

$$|\lambda E - B| = |\lambda E - P^{-1}AP|$$

$$= |\lambda P^{-1}EP - P^{-1}AP|$$

$$= |P^{-1}||\lambda E - A||P|$$

$$= |\lambda E - A|$$

$\therefore A, B$ 有相同的特征值.

$$\because \sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii} \quad \therefore \text{tr}(A) = \text{tr}(B)$$

$$\lambda_1 \lambda_2 \cdots \lambda_n = |A| \quad \therefore |A| = |B|$$

* 有相同特征值未必相似

2) $A \sim B$ A 可逆的充要条件为 B 可逆

若 A, B 均可逆 则 $A^{-1} \sim B^{-1}$

$$\because A \sim B \quad \therefore |A| = |B| \quad \begin{cases} \neq 0 & A, B \text{ 可逆} \\ = 0 & A, B \text{ 不可逆} \end{cases}$$

* $A \sim B$, A 与 B 同时可逆或同时不可逆

$$\because P^{-1}AP = B \quad \therefore B^{-1} = (P^{-1}AP)^{-1}$$

$$\therefore B^{-1} = P^{-1}A^{-1}P \quad \therefore A^{-1} \sim B^{-1}$$

3) 若 $A \sim B$ 则 $A^m \sim B^m$

$$\because P^{-1}AP = B \quad \therefore B^m = (P^{-1}AP)^m \\ = P^{-1}A^mP$$



例 1) $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & a \end{bmatrix}$ $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 2 \\ 0 & -1 & b \end{bmatrix}$

$A \sim B$ 求 a, b

$A \sim B$ $\begin{cases} \text{tr}(A) = \text{tr}(B) \\ |A| = |B| \\ \text{均可逆或均不可逆} \\ A^{-1} \sim B^{-1} \\ A^m \sim B^m \\ \text{特征值相同} \\ r(A) = r(B) \end{cases}$

$\begin{cases} 2+1+a = 1+3+b \\ \begin{vmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & a \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 3 & 2 \\ 0 & -1 & b \end{vmatrix} \end{cases}$

$P^{-1}AP = \Lambda$ 对角形.

与对角形相似的条件.

定理: A 相似于 $\Lambda \Leftrightarrow A$ 有 n 个线性无关的特征向量.

$\Rightarrow$ 可逆 P $P^{-1}AP = \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}$

设 $P = (\alpha_1 \dots \alpha_n)$

$P^{-1}AP = \Lambda$ $AP = P\Lambda$

$A(\alpha_1, \alpha_2, \dots, \alpha_n) = (\alpha_1, \alpha_2, \dots, \alpha_n) \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}$

$(A\alpha_1, A\alpha_2, \dots, A\alpha_n) = (\lambda_1\alpha_1, \lambda_2\alpha_2, \dots, \lambda_n\alpha_n)$

$A\alpha_1 = \lambda_1\alpha_1$ $A\alpha_2 = \lambda_2\alpha_2$ $\dots$ $A\alpha_n = \lambda_n\alpha_n$

α_i : P 为可逆矩阵, α_i 为 P 的列向量

$\therefore \alpha_i \neq 0 \therefore A\alpha_i = \lambda_i\alpha_i$

$\therefore r(P) = n$ $r(\alpha_1, \alpha_2, \dots, \alpha_n) = n$

$\therefore \alpha_1, \alpha_2, \dots, \alpha_n$ 线性无关.

(充分条件)

推论: A 有 n 个互异的特征根(值).

例 $A \sim \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}$

$\lambda_1, \lambda_2, \dots, \lambda_n$ 互异

$\alpha_1, \alpha_2, \dots, \alpha_n$ 无关

即 n 个线性无关的特征向量.

$\therefore A \sim \Lambda$ (由定理)

A 为 6 阶方阵

① 若求出的特征根均为单根.

则一定相似于对角形.

1) $\lambda_1 = 1$ $\lambda_2 = 2$ $\lambda_3 = 3$ $\lambda_4 = 4$ $\lambda_5 = 5$ $\lambda_6 = 6$

② 有单根, 也有重根.

2) $\lambda_1 = 1$ $\lambda_2 = \lambda_3 = 5$ $\lambda_4 = \lambda_5 = \lambda_6 = 8$

6 阶方阵 $\rightarrow$ 6 个线性无关特征向量
线性无关特征向量不会超过其重数

$\lambda_1 = 1$ $\lambda_2 = \lambda_3 = 5$ $\lambda_4 = \lambda_5 = \lambda_6 = 8$

$\downarrow$ 单根 二重 三重

1个 $(\)$ $(\)$ $(\)$

* (重根 线性无关的特征向量的个数) 不超过其重数



扫描全能王 创建

当重根对应线性无关的特征向量的个数 = 重数 时有 n 个线性无关特征向量。

例: $A = \begin{pmatrix} 3 & 2 & -1 \\ -2 & -2 & 2 \\ 3 & 6 & -1 \end{pmatrix}$ 相似于对角形? $P = ? \quad \Lambda = ?$

解: $|\lambda E - A| = \begin{vmatrix} \lambda-3 & -2 & 1 \\ 2 & \lambda+2 & -2 \\ -3 & -6 & \lambda+1 \end{vmatrix}$

$$= \begin{vmatrix} \lambda-2 & -2 & 1 \\ 0 & \lambda+2 & -2 \\ \lambda-2 & -6 & \lambda+1 \end{vmatrix}$$

$$= (\lambda-2)^2(\lambda+4) = 0$$

$$\lambda_1 = -4 \quad \lambda_2 = \lambda_3 = 2 \quad (\text{二重})$$

对于 $(\lambda E - A)X = 0$

① 当 $\lambda_1 = -4$ $(\lambda_1 E - A)X = 0$ $\lambda_1 = -4$ 代入第一个行列式

$$\begin{pmatrix} -7 & -2 & 1 \\ 2 & -2 & -2 \\ -3 & -6 & -3 \end{pmatrix} \rightarrow \text{行} \rightarrow \begin{pmatrix} 1 & 0 & -\frac{1}{3} \\ 0 & 1 & \frac{2}{3} \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{cases} x_1 = \frac{1}{3}x_3 \\ x_2 = -\frac{2}{3}x_3 \end{cases} \quad x_3 \text{ 为自由未知量.}$$

基础解系 $\alpha_1 = \begin{pmatrix} \frac{1}{3} \\ -\frac{2}{3} \\ 1 \end{pmatrix}$

② $\lambda_2 = \lambda_3 = 2$ 时

基础解系

$$\alpha_2 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \quad \alpha_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

重根基础解系向量个数 必须与重数相同才相似于对角形。

$P = (\alpha_1, \alpha_2, \alpha_3)$ 若 P 为 $(\alpha_3, \alpha_1, \alpha_2)$

$$\Lambda = \begin{pmatrix} -4 & & \\ & 2 & \\ & & 2 \end{pmatrix} \quad \Lambda = \begin{pmatrix} 2 & & \\ & -4 & \\ & & 2 \end{pmatrix}$$

“对应”

例: A 为三阶. $\lambda_1 = \lambda_2 = 1 \rightarrow \alpha_1$

$$\lambda_3 = 2 \rightarrow \alpha_2$$

二重根但基础解系中仅一个向量

$\Rightarrow$ 不与对角形相似

定理: $A \sim \Lambda$

$\Leftrightarrow r_i$ 重特征根, 基础解系

有 r_i 个解, $n - r(\lambda_i E - A) = r_i$

(* 齐次线性方程组基础解系解的个数) 为 $n - r_i$

$$n - r_i = r(\lambda_i E - A)$$

每个系数矩阵的秩为 $n - r_i$



例5): 3阶方阵A. $\lambda_1=1$ $\lambda_2=2$ $\lambda_3=3$. 例6)

对应特征向量 $\alpha_1=(1, 1, 1)^T$

$\alpha_2=(1, 2, 3)^T$ $\alpha_3=(1, 3, 6)^T$

1) 求矩阵A. 2) A^T 的特征值与特征向量

$|\lambda E - A| = 0$ 解方程.

解: $P=(\alpha_1, \alpha_2, \alpha_3)$

$$P^{-1}AP = \begin{pmatrix} 1 & & \\ & 2 & \\ & & 3 \end{pmatrix} = \Lambda$$

$$A = P\Lambda P^{-1}. \quad A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & -1 & 2 \\ 3 & -9 & 7 \end{pmatrix}$$

12) A^T 的特征值与A同

$\lambda_1=1$ $\lambda_2=2$ $\lambda_3=3$.

$$P^{-1}AP = \Lambda \quad (P^{-1}AP)^T = P^T A^T (P^{-1})^T$$

$$\therefore P^{-1}AP = \Lambda \quad \therefore (P^{-1}AP)^T = \Lambda$$

$$\therefore P^T A^T (P^{-1})^T = \Lambda$$

$$P^T A^T (P^T)^{-1} = \Lambda$$

$$[(P^T)^{-1}]^{-1} A^T (P^T)^{-1} = \Lambda$$

$$(P^T)^{-1} = \begin{pmatrix} 3 & -3 & 1 \\ -3 & 5 & -2 \\ 1 & -2 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 0 & 1 & 1 \\ -2 & 0 & 2 \\ -1 & 1 & 0 \end{pmatrix} \quad A^{100}$$

解: $\lambda_1=-1$ $\lambda_2=0$ $\lambda_3=1$

由于均为单根, $\therefore$ 必有与之相似的对角阵.

$$\text{基础解系: } \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}$$

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 4 \\ 1 & 1 & 3 \end{pmatrix} \quad P^{-1}AP = \Lambda$$

$$A = P\Lambda P^{-1}$$

$$A^{100} = P\Lambda^{100}P^{-1}$$

$$\Lambda = \begin{pmatrix} -1 & & \\ & 0 & \\ & & 1 \end{pmatrix}$$

5.3 实对称矩阵的对角化

5.2 有n个线性无关的特征向量的矩阵

才能对角化 $P^{-1}AP = \Lambda$.

* 所有实对称矩阵均能对角化.

$$\text{内积: } \begin{pmatrix} \alpha \\ 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} \beta \\ 0 \\ 0 \\ 8 \end{pmatrix} \quad \text{分量对应相乘再相加}$$

$$(\alpha, \beta) = 1 \times 0 + 2 \times 0 + 3 \times 8 = 24$$



$$\alpha = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \quad \beta = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

$$(\alpha, \beta) = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

其内积是一个数。

若 α, β 均为列向量, $(\alpha, \beta) = \alpha^T \beta$

$(\alpha, \beta) = \alpha \beta^T$ α, β 均为行向量。

$$① (\alpha, \alpha) \geq 0 \quad (\alpha, \alpha) = a_1^2 + a_2^2 + \dots + a_n^2 \geq 0$$

$(\alpha, \alpha) = 0$ 当且仅当 α 为零向量。

$$② (\alpha, \beta) = (\beta, \alpha)$$

$$③ (k\alpha, \beta) = k(\alpha, \beta) = (\alpha, k\beta)$$

$$(k\alpha, k\beta) = k^2(\alpha, \beta)$$

$$④ (\alpha + \beta, \gamma) = (\alpha, \gamma) + (\beta, \gamma)$$

$$(\alpha, \beta + \gamma) = (\alpha, \beta) + (\alpha, \gamma)$$

$$(k_1 \alpha + k_2 \beta, \gamma) = k_1(\alpha, \gamma) + k_2(\beta, \gamma)$$

$$(\alpha, k_1 \beta + k_2 \gamma) = k_1(\alpha, \beta) + k_2(\alpha, \gamma)$$

$$(k_1 \alpha_1 + k_2 \alpha_2, m_1 \beta_1 + m_2 \beta_2)$$

$$= k_1 m_1 (\alpha_1, \beta_1) + k_1 m_2 (\alpha_1, \beta_2) + k_2 m_1 (\alpha_2, \beta_1) + k_2 m_2 (\alpha_2, \beta_2)$$

向量的长度 (范数, 模)

$$\|\alpha\| = \sqrt{(\alpha, \alpha)} \quad (\alpha, \alpha) = \|\alpha\|^2$$

$$\alpha = (-1, -1, 5) \quad \|\alpha\| = \sqrt{(-1)^2 + (-1)^2 + 5^2}$$

点到原点的距离。

$\|\alpha\| = 1$ 单位向量。

$$\alpha = (1, 0, 0)$$

非单位向量可以进行单位化 (标准化)

$$\frac{\alpha}{\|\alpha\|} = \beta, \quad \beta \text{ 为单位向量。}$$

$$\|\beta\| = \frac{1}{\|\alpha\|} \sqrt{(\frac{1}{\|\alpha\|} \alpha, \frac{1}{\|\alpha\|} \alpha)}$$

$$= \sqrt{\frac{1}{\|\alpha\|^2} (\alpha, \alpha)}$$

$$= \frac{1}{\|\alpha\|} \sqrt{(\alpha, \alpha)}$$

$$= 1$$

性质: 1) $\|\alpha\| \geq 0$

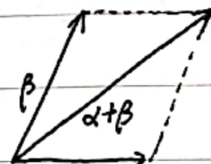
$$\|\alpha\| = 0 \iff \alpha = 0$$

$$2) \|k\alpha\| = |k| \|\alpha\|$$

$$\|k\alpha\| = \sqrt{(k\alpha, k\alpha)} = \sqrt{k^2 (\alpha, \alpha)} = |k| \|\alpha\|$$

$$3) |(\alpha, \beta)| \leq \|\alpha\| \cdot \|\beta\|$$

$$|a_1 b_1 + a_2 b_2 + \dots + a_n b_n| \leq \sqrt{a_1^2 + a_2^2 + \dots + a_n^2} \sqrt{b_1^2 + \dots + b_n^2}$$



$$4) \|\alpha + \beta\| \leq \|\alpha\| + \|\beta\| \quad (\text{三角})$$



正交(垂直) $(\alpha, \beta) = 0 \quad \alpha \perp \beta$

零向量和任何向量都正交 $(0, \alpha) = 0$

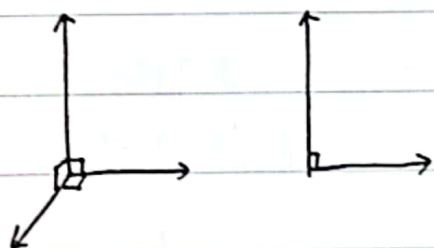
$$(\alpha, \alpha) = 0 \Leftrightarrow \alpha = 0$$

定义: $\alpha_1, \alpha_2, \dots, \alpha_s$ 两两正交, 则称为
正交向量组. (正交向量组不含零
向量)

如果每个向量都是单位向量, 则为标

标准正交向量组 $\begin{cases} (\alpha_i, \alpha_i) = 1 \\ (\alpha_i, \alpha_j) = 0 \end{cases}$

正交(垂直)



定理: $\alpha_1, \dots, \alpha_s$ 为正交向量组, $\alpha_1, \dots, \alpha_s$
线性无关.

证明: 假设 $k_1 \alpha_1 + k_2 \alpha_2 + \dots + k_s \alpha_s = 0$

$$(\alpha_1, k_1 \alpha_1 + k_2 \alpha_2 + \dots + k_s \alpha_s) = (\alpha_1, 0) = 0$$

$$\therefore k_1 (\alpha_1, \alpha_1) + k_2 (\alpha_1, \alpha_2) + \dots + k_s (\alpha_1, \alpha_s) = 0$$

$$\because (\alpha_1, \alpha_2) = 0, (\alpha_1, \alpha_3) = \dots = (\alpha_1, \alpha_s) = 0$$

$$\therefore k_1 (\alpha_1, \alpha_1) = 0 \quad \text{又} \because (\alpha_1, \alpha_1) \neq 0$$

$$\therefore k_1 = 0$$

再分别与 $\alpha_2, \alpha_3, \dots, \alpha_s$ 作内积

同理. 可知 $k_1 = k_2 = \dots = k_s = 0$

$\therefore$ 线性无关

施密特正交化

给一组线性无关的 $\alpha_1, \alpha_2, \dots, \alpha_s$

求正交的 $\beta_1, \beta_2, \dots, \beta_s$ 且与之等价.

$$\beta_1 = \alpha_1$$

$$\beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)} \beta_1$$

$$\beta_3 = \alpha_3 - \frac{(\alpha_3, \beta_1)}{(\beta_1, \beta_1)} \beta_1 - \frac{(\alpha_3, \beta_2)}{(\beta_2, \beta_2)} \beta_2$$

$$\text{单位化: } e_1 = \frac{\beta_1}{\|\beta_1\|} \quad e_2 = \frac{\beta_2}{\|\beta_2\|} \quad \dots$$

$$\alpha_1, \alpha_2, \dots, \alpha_s \xrightarrow{\text{正交化}} \beta_1, \dots, \beta_s \xrightarrow{\text{单位化}} \eta_1, \eta_2, \dots, \eta_s$$

$$\text{例: } \alpha_1 = (1, 1, 1, 1)^T \quad \alpha_2 = (1, -2, -3, -4)^T \\ \alpha_3 = (1, 2, 2, 3)^T$$

$$\beta_1 = \alpha_1 \quad \beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)} \beta_1$$

$$\beta_3 = \alpha_3 - \frac{(\alpha_3, \beta_1)}{(\beta_1, \beta_1)} \beta_1 - \frac{(\alpha_3, \beta_2)}{(\beta_2, \beta_2)} \beta_2$$

$$\eta_1 = \frac{\beta_1}{\|\beta_1\|} \quad \eta_2 = \frac{\beta_2}{\|\beta_2\|} \quad \eta_3 = \frac{\beta_3}{\|\beta_3\|}$$

正交矩阵: A 为 n 阶方阵. $A^T A = E$

性质1: A 是正交矩阵 $|A| = 1$ or -1

$$|A^T| |A| = 1 \quad |A|^2 = 1 \quad \therefore |A| = 1 \text{ 或 } -1$$

2): A 为正交矩阵 $A^T = A^{-1}$, 且

A^{-1} 和 A^T 均为正交矩阵.



$$\boxed{A^T A = E}$$

$$\because A^T A = E \quad \therefore A^{-1} = A^T$$

$$(A^{-1})^T A^{-1} = (A^T)^T A^T = (A A^{-1})^T = E \quad \sqrt{(\alpha, \alpha)} = \|\alpha\|$$

$$(A^T)^T A^T = A A^T = A A^{-1} = E$$

3) A, B 为 n 阶正交矩阵, AB 也正交

$$(AB)^T AB = B^T A^T AB = B^T E B = E$$

4) A 为正交矩阵, α, β 为 n 维列向量

$$(\alpha, \beta) = (\alpha, \beta)$$

$$(\alpha, \beta) = \alpha^T \beta \text{ 或 } \alpha \beta^T \quad \left(\begin{array}{c} \\ \end{array} \right)$$

$$(\alpha, \beta) = (\alpha)^T A \beta = \alpha^T A^T A \beta = (\alpha, \beta)$$

定理: A 正交 $\Leftrightarrow A$ 的列向量组/行向量组

是标准正交向量组

$$\text{证: } A = (\alpha_1, \alpha_2, \dots, \alpha_n)$$

$$A^T A = (\alpha_1, \alpha_2, \dots, \alpha_n)^T (\alpha_1, \alpha_2, \dots, \alpha_n)$$

$$= \begin{pmatrix} \alpha_1^T \\ \alpha_2^T \\ \vdots \\ \alpha_n^T \end{pmatrix} (\alpha_1, \alpha_2, \dots, \alpha_n)$$

$$= \begin{pmatrix} \alpha_1^T \alpha_1 & \alpha_1^T \alpha_2 & \dots & \alpha_1^T \alpha_n \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_n^T \alpha_1 & \alpha_n^T \alpha_2 & \dots & \alpha_n^T \alpha_n \end{pmatrix}$$

$$= \begin{pmatrix} (\alpha_1, \alpha_1) & \dots & (\alpha_1, \alpha_n) \\ \vdots & \ddots & \vdots \\ (\alpha_n, \alpha_1) & \dots & (\alpha_n, \alpha_n) \end{pmatrix}$$

$$\underline{A^T A = E} = \begin{pmatrix} 1 & & \\ & 1 & \\ & & \ddots \\ & & & 1 \end{pmatrix}$$

$$\therefore (\alpha_i, \alpha_i) = 1 = \|\alpha_i\|^2 = 1 \quad (\alpha_i, \alpha_j) = 0$$

标准, 正交

$$* : (\alpha, \alpha) = \|\alpha\|^2$$

$$\sqrt{(\alpha, \alpha)} = \|\alpha\|$$

例2: 三阶非零矩阵 A .

$a_{ij} = A_{ij}$ 求证 $|A|=1$ 且 A 正交.

证明:

$$A^* = \begin{pmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{pmatrix}$$

$$= \begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{pmatrix} = A^T$$

$$\therefore A^* = A^T$$

$$A^T A = A^* A = |A| E$$

$$|A^T| |A| = |A|^3 \quad |A|^2 = |A|^3$$

$$|A|^2 (|A| - 1) = 0 \quad \text{下证 } |A| \neq 0$$

设 $a_{11} \neq 0$ (A 为 3 阶非零)

$$|A| = a_{11} A_{11} + a_{12} A_{12} + a_{13} A_{13} \quad (\text{按行展开})$$

$$= a_{11}^2 + a_{12}^2 + a_{13}^2 > 0$$

$$\therefore |A| \neq 0 \quad \therefore |A| = 1$$

★ 实对称矩阵的对角化

n 阶实对称矩阵, 它的 n 个特征值都是实数, 并且它的特征向量都是单位向量.



扫描全能王 创建

定理: 实对称矩阵 A 的不同特征值对应特征向量正交.

$$A^T = A$$

$$\lambda_1, \lambda_2 (\lambda_1 \neq \lambda_2)$$

$$\downarrow \quad \downarrow$$

$$\text{证明: } (\alpha_1, \alpha_2) = 0$$

$$A\alpha_1 = \lambda_1 \alpha_1 \quad A\alpha_2 = \lambda_2 \alpha_2$$

$$(A\alpha_1, \alpha_2) = (\lambda_1 \alpha_1, \alpha_2) = \lambda_1 (\alpha_1, \alpha_2)$$

$$= \lambda_1 \alpha_1^T \alpha_2$$

$$(A\alpha_1, \alpha_2) = (A\alpha_1)^T \alpha_2 = \alpha_1^T A^T \alpha_2$$

$$\because A^T = A \therefore (A\alpha_1, \alpha_2) = \alpha_1^T A \alpha_2$$

$$= \lambda_2 \alpha_1^T \alpha_2 = \lambda_2 (\alpha_1, \alpha_2)$$

$$(\lambda_1 - \lambda_2) (\alpha_1, \alpha_2) = 0 \quad \because (\lambda_1 \neq \lambda_2)$$

$$\therefore (\alpha_1, \alpha_2) = 0$$

推论: 设 A 为 n 阶实对称矩阵.

λ 是 A 的特征方程的 k 重根.

则矩阵 $A - \lambda E$ 的秩 $R(A - \lambda E) = n - k$, 从而对应特征值 λ 恰有 k 个线性无关的特征向量.

给定实对称矩阵 A , 求正交矩阵 Q

使 $Q^T A Q = \Lambda$

矩阵的对角化?

① 如果有 n 个线性无关的特征向量

特征向量作成列. $P^T A P = \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{pmatrix}$

② 如果没有 n 个线性无关的特征向量 则不能对角化.

正交相似: A, B 同阶. 存在正交矩阵 P

$$\text{使 } P^T A P = B.$$

相似: A, B 同阶. 可逆矩阵 P

$$\text{使 } P^{-1} A P = B.$$

实对称矩阵一定能对角化

S_1 : 求特征值 S_2 : 求特征向量

$\star S_3$: 特征向量正交化. 单位化

S_4 : 处理后的特征向量作成列

构成 Q

定理: A 为实对称矩阵. 一定存在正交矩阵 Q

$$\text{使 } Q^T A Q = \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{pmatrix}$$

$$S_5: \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{pmatrix} \quad \text{注意顺序}$$

n 阶实对称矩阵 A , 一定有 n 个线性无关的特征向量.



A为3阶实对称矩阵.

$$1) \lambda_1 = 2 \quad \lambda_2 = 5 \quad \lambda_3 = 7$$

$\alpha_1 \quad \alpha_2 \quad \alpha_3$

三个根都是单根.

由于实对称矩阵不同的特征值对应的特征向量一定正交, 故无需作施密特正交化, 直接单位化即可

$$\eta_1 = \frac{\alpha_1}{\|\alpha_1\|} \quad \eta_2 = \frac{\alpha_2}{\|\alpha_2\|} \quad \eta_3 = \frac{\alpha_3}{\|\alpha_3\|}$$

$$Q = (\eta_1, \eta_2, \eta_3) \quad Q^T A Q = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{pmatrix}$$

$$2) \lambda_1 = 1 \text{ (单)} \quad \lambda_2 = \lambda_3 = 10 \text{ (二重根)}$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ \alpha_1 & \xrightarrow{\text{正交}} & \alpha_2, \alpha_3 \\ & \xrightarrow{\text{正交}} & \end{array}$$

只需对 α_2, α_3 进行施密特正交化

$$\alpha_2, \alpha_3 \rightarrow \beta_2, \beta_3 \quad \beta_1 = \alpha_1$$

$$\beta_1, \beta_2, \beta_3 \xrightarrow{\text{单位化}} \eta_1, \eta_2, \eta_3$$

$$3) \lambda_1 = \lambda_2 = \lambda_3 = 9 \text{ (三重)}$$

$$\alpha_1, \alpha_2, \alpha_3 \xrightarrow{\text{正交化}} \xrightarrow{\text{单位化}} \eta_1, \eta_2, \eta_3$$

$$\text{例: } A = \begin{pmatrix} 4 & -4 & 2 \\ -4 & 4 & -2 \\ 2 & -2 & 1 \end{pmatrix} \quad \text{对称.}$$

$$1) |\lambda E - A| = \begin{vmatrix} \lambda - 4 & 4 & -2 \\ 4 & \lambda - 4 & 2 \\ -2 & 2 & \lambda - 1 \end{vmatrix}$$

$$= \lambda^2(\lambda - 9) = 0$$

$$\therefore \lambda_1 = \lambda_2 = 0, \quad \lambda_3 = 9$$

$$\text{对 } \lambda_1 = \lambda_2 = 0 \text{ (求 } \alpha_1, \alpha_2) \rightarrow \eta_1, \eta_2$$

$$\text{对 } \lambda_3 = 9 \text{ (求 } \alpha_3) \xrightarrow{\text{单位化}} \eta_3$$

$$Q = (\eta_1, \eta_2, \eta_3) \quad \Lambda = \begin{pmatrix} 0 & & \\ & 0 & \\ & & 9 \end{pmatrix}$$

例: 三阶实对称矩阵 A.

特征值为 6, 3, 3.

$$6 \text{ 对应的特征向量 } \alpha_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

解: 3 对应的特征向量 $\alpha = (x_1, x_2, x_3)^T$

$$\because \text{正交} \therefore (\alpha, \alpha_1) = (x_1 + x_2 + x_3) = 0$$

$$x_1 = -x_2 - x_3, \quad x_2, x_3 \text{ 自由.}$$

$$\alpha_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad \alpha_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \quad P = (\alpha_1, \alpha_2, \alpha_3)$$

$$P^T A P = \Lambda \quad A = P \Lambda P^{-1}$$

$$= \begin{pmatrix} & & \\ & 6 & \\ & & 3 \end{pmatrix} \begin{pmatrix} & & \\ & & \\ & & 3 \end{pmatrix} \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

